Data for Policy Analysis and Management

CLASS 4: BIVARIATE DESCRIPTIVE STATISTICS AND BASIC STATISTICAL TESTS

Aida Pacheco-Applegate

Summer 2026

Agenda

1. Bivariate distributions and relationships

2. Types of Bivariate Analysis and Display

3. Statistical Difference and Tests:
   - Two-sample t-test
   - Two-sample z-test
   - ANOVA
   - Chi-square test
   - Paired test

4. Key Concepts to Understand R Outputs

5. Lab Session: Relationships and Tests

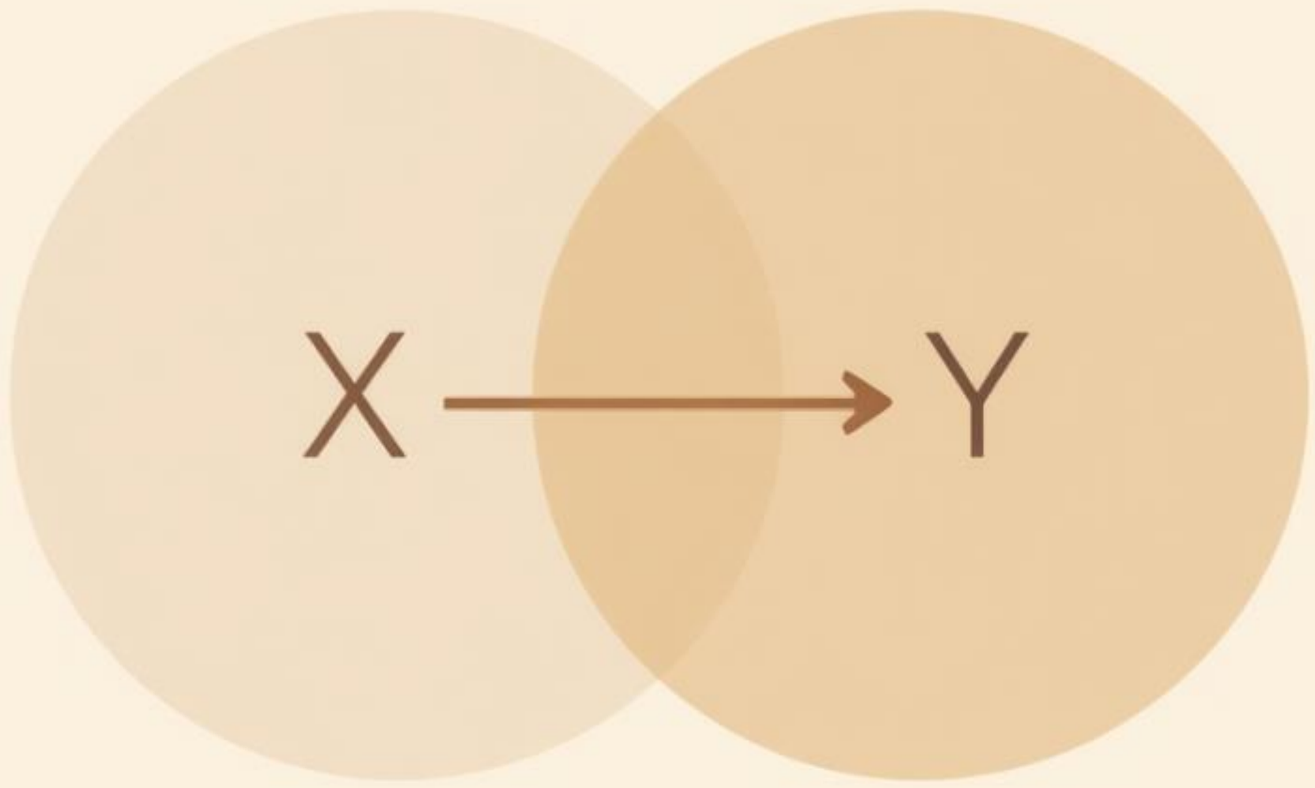
Bivariate Distributions & Relationships

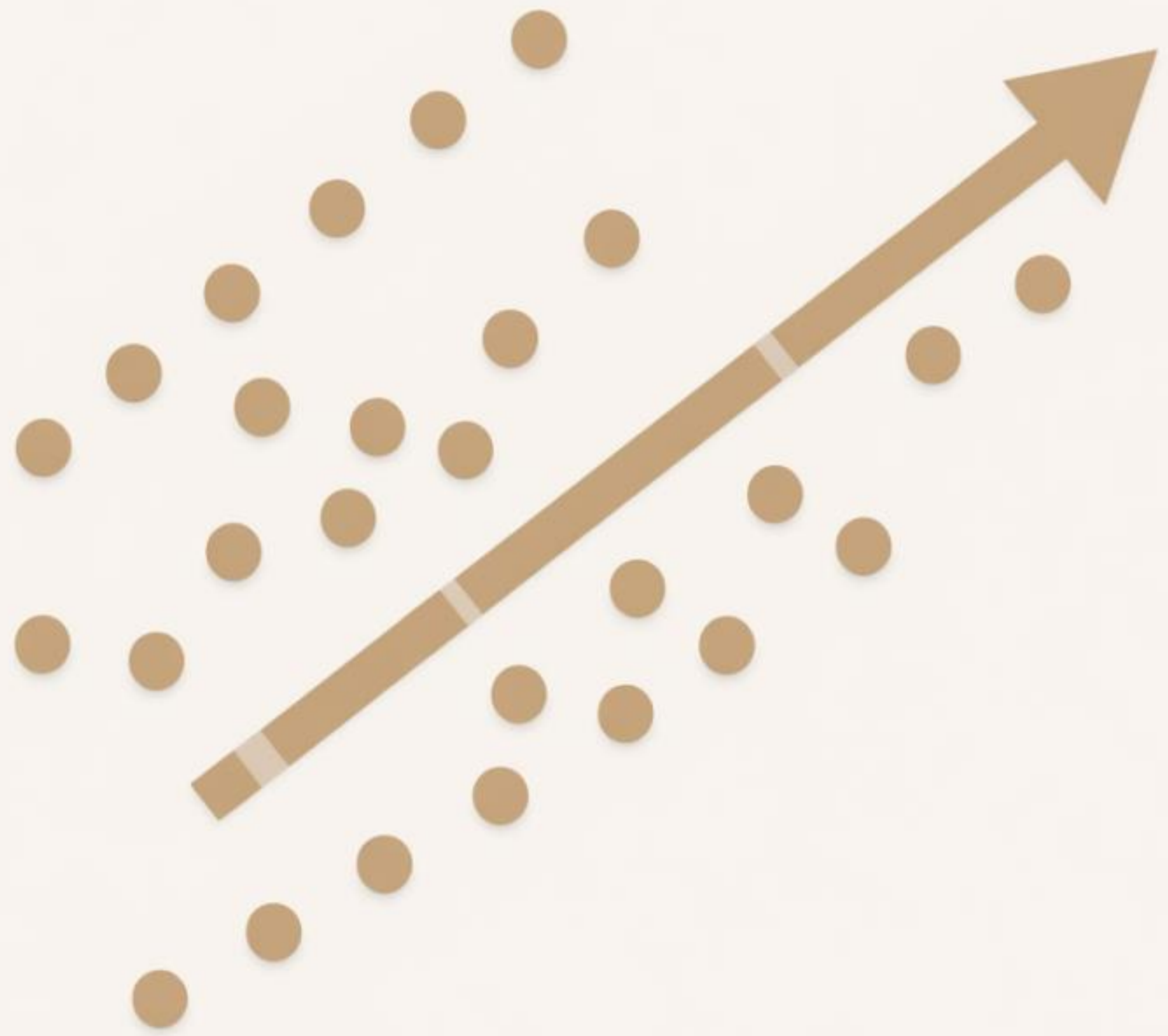
Core Idea

- Most research questions focus on relationships between two variables, not just a single variable.
- Bivariate analysis describes, compares, and interprets associations between them.

Key Concepts

- We move from univariate to bivariate analysis
- Core question: "What does knowing X tell us about Y?"
- Response variable (Y) = outcome of interest
- Explanatory variable (X) = grouping or predictor





What Is a Relationship?

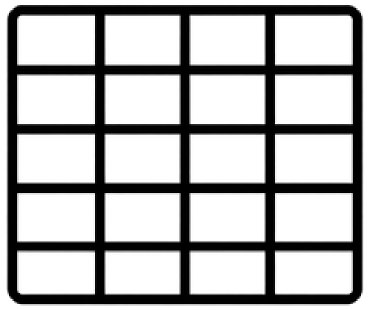
Definition

- A relationship between two variables describes how they co-vary: when values of one variable change, values of the other tend to change in a predictable way.

Types of Relationships

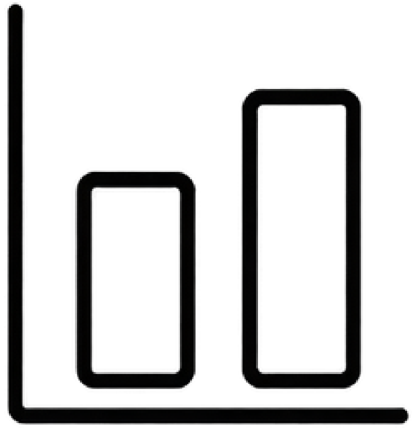
- Positive: as X increases, Y increases (e.g., age & height in children)
- Negative: as X increases, Y decreases (e.g., age & frequency of crying)
- Null: no consistent pattern between X and Y
- Association: distributions or outcomes shift across values of another variable

Types of Bivariate Analysis



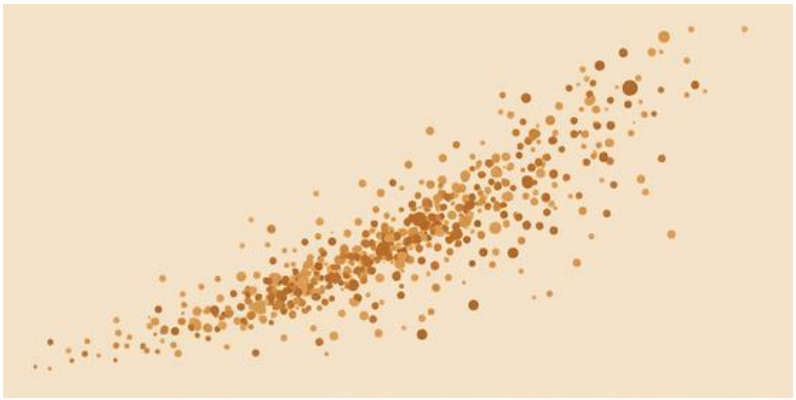
Discrete × Discrete

- Compare proportions across groups using contingency tables.
- Example: % of adults taking vitamin E by smoking status.



Discrete × Continuous

- Compare conditional means across groups using means tables.
- Example: average income by gender.



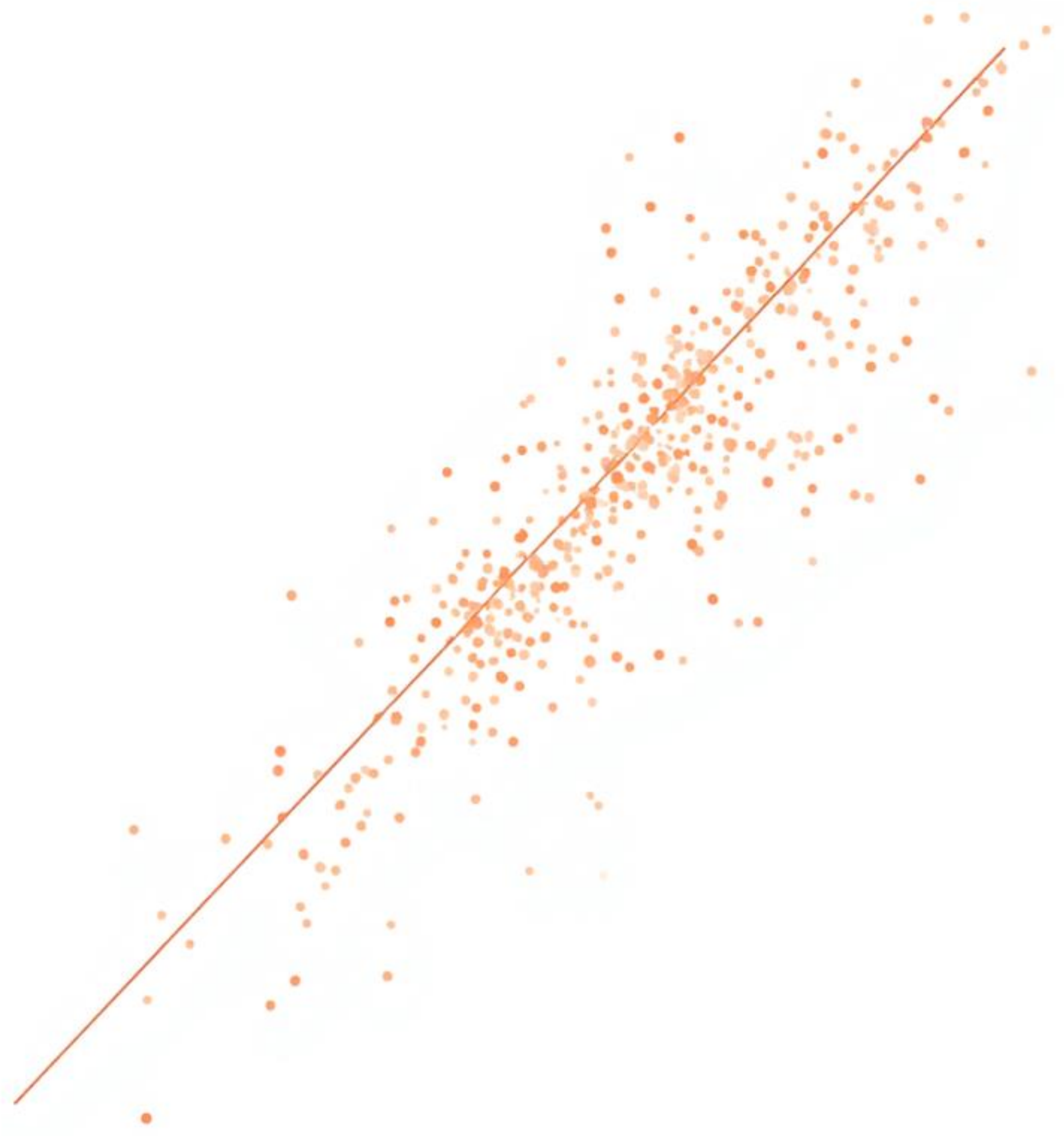
Continuous × Continuous

- Examine continuous relationships / correlation or regression coefficients.
- Example: education level and earnings.

Scatterplots (Continuous Variables)

- A scatterplot displays the relationship between two continuous variables by placing each observation as a single point defined by its (X, Y) coordinates.
- Key features to examine:
  - Direction — positive (↑X → ↑Y) or negative (↑X → ↓Y)
  - Strength — how tightly clustered the points are around a trend
  - Shape — linear, non-linear, or no discernible pattern

Scatterplots are the ***first step*** *in understanding associations in numerical data, and serve as the foundation for correlation and regression analysis.*



Correlation (Continuous Variables)

What Is Correlation?

- Correlation quantifies the linear relationship between two quantitative variables.
- It tells us the direction and strength of the association (but only for linear patterns!)

Key Properties

- Range: −1 to +1 (closer to ±1 = stronger relationship)
- Positive correlation: both variables increase together (↑X → ↑Y)
- Negative correlation: one increases as the other decreases (↑X → ↓Y)
- Value near 0: little to no linear association
- Only measures linear association (not all patterns!)

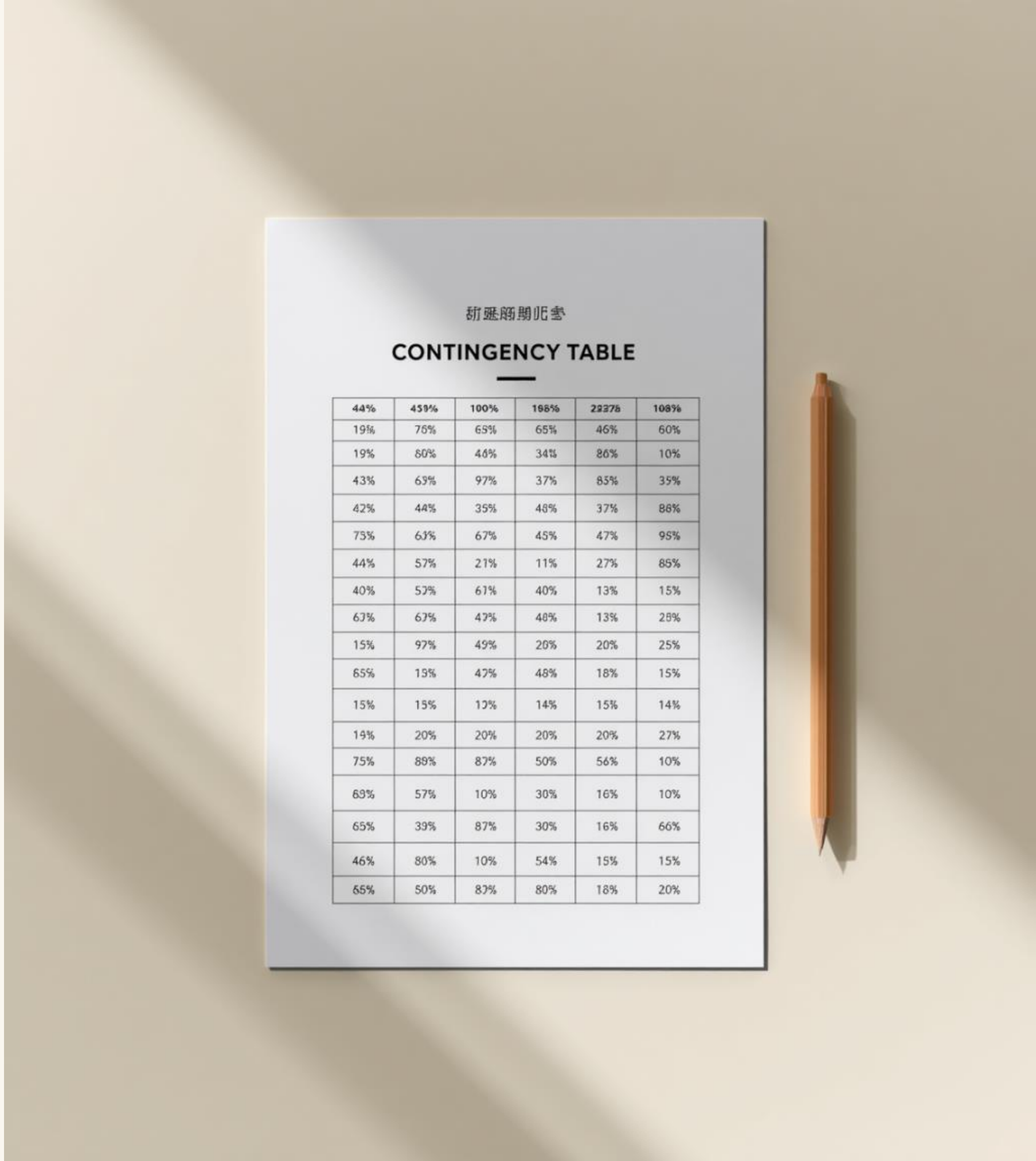
Conditional Distributions (Discrete Variables)

Key Concept

- For discrete variables, use contingency tables instead of scatterplots.
- Compare conditional distributions using row or column percentages to identify associations.

How to Interpret

- Similar percentages across groups → weak or no relationship.
- Different percentages across groups → evidence of association.
- Example: % choosing center-based childcare by race/ethnicity



What Is Statistical Difference?

Definition

- A statistical difference is a difference between groups that is unlikely to be explained by chance alone.
- It suggests a real pattern in the population, not just random variation in the sample.
- Statistical tests help us determine whether an observed difference is meaningful or simply due to sampling variability.

Example: Income by Gender

X-ray Technicians' Average Income:

- Women: $50,000
- Men: $55,000

Is this $5,000 difference real or just random variation from sampling?

- A statistical test answers exactly this question. It evaluates whether the observed difference is larger than what we would expect by chance, helping us make evidence-based conclusions from data.

Statistical Tests

Purpose & Role

- Statistical tests help us determine whether the observed pattern is large enough to provide evidence of a real effect.

What Tests Do

- Evaluate evidence for or against a hypothesis
- Determine if observed differences are likely due to chance
- Make inferences from a sample to the broader population
- Support evidence-based policy and research decisions
- *Note: Tests measure statistical significance — NOT the importance or size of an effect.*

Statistical Tests for Relationships Between Two Variables

- Most applied research asks a simple question: Do two groups differ?
- We answer this by comparing outcomes across groups defined by a second variable.
- The key idea is bivariate analysis:
- One variable defines the groups (explanatory variable) → X
- One variable is the outcome we measure (response variable) → Y

Discrete Outcome

- Used when the response variable (Y) is categorical or binary.
- Core question: Do two groups differ *in the probability* of an outcome?
- Example: Is the probability of actively searching for a job different between men and women?
- Typical tests: Two-sample z-test (proportions), Chi-square test.

Continuous Outcome

- Used when the response variable (Y) is numerical or quantitative.
- Core question: *Do two groups differ in their average value on an outcome?*
- Example: Does average income differ by age group?
- Typical tests: Two-sample t-test (independent means), Paired t-test (dependent samples).

Most Common Tests

Two-Sample t-test

Compare means between two independent groups (e.g., average income by gender).

Two-Sample z-test

Compare proportions between two independent groups (e.g., AI tool use: adolescents vs. young adults).

Chi-Square Test

Compare categorical variables across groups (e.g., childcare type by race).

One-way ANOVA

Compare means across multiple groups (e.g., income by race).

Independent samples

Paired t-test

Compare same individuals measured twice or matched pairs (e.g., test scores before and after tutoring).

Non-independent samples

Independent Samples

Most Common Case

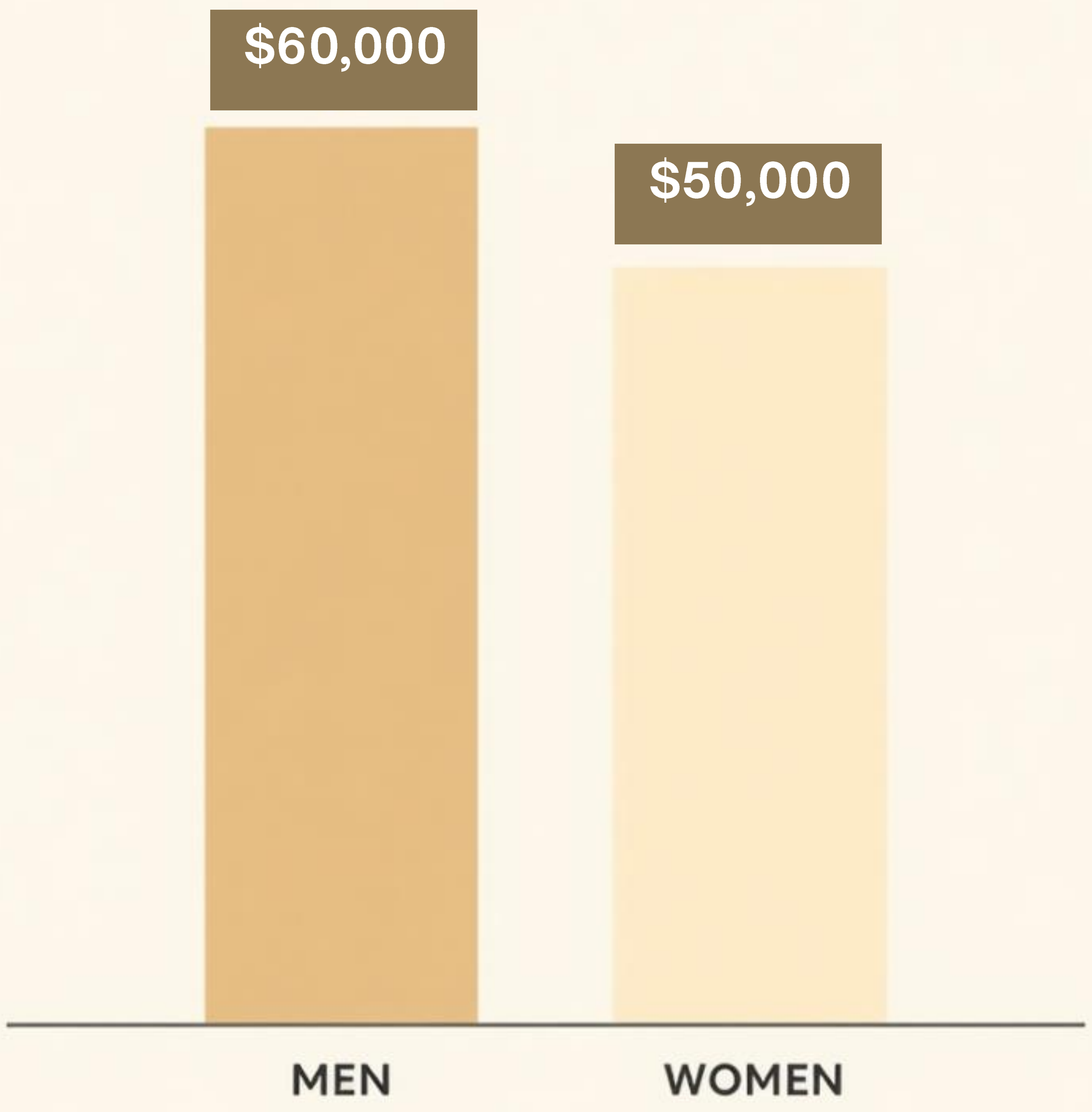
- Independent samples consist of two groups made up of entirely different people.
- The groups are unrelated to each other, and their variation is separate and distinct.

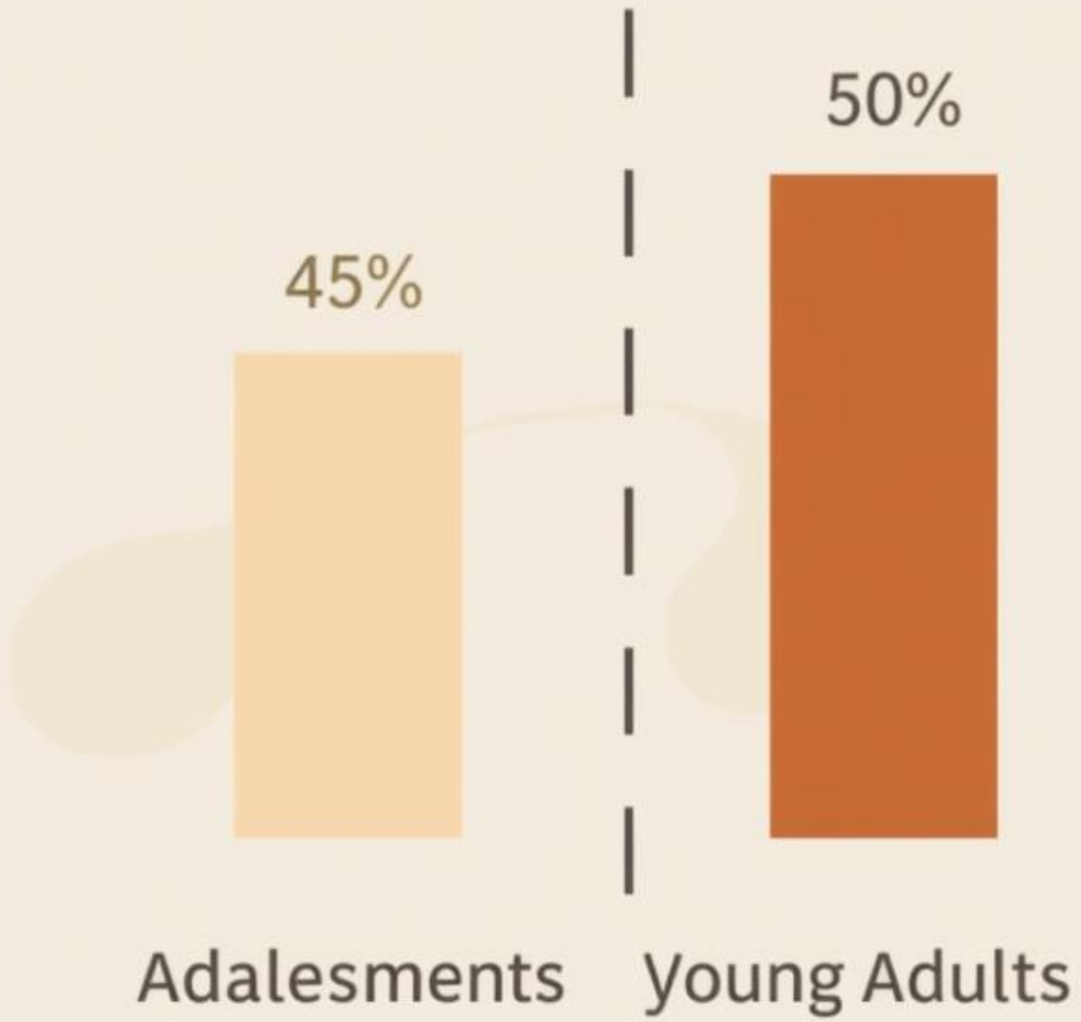
Key Characteristics

- Groups are composed of different individuals (e.g., men vs. women, treatment vs. control)
- No relationship or connection between members of each group
- Variation within each group is independent of the other
- Common tests: two-sample t-test (comparing means) and two-sample z-test (comparing proportions)

Two-Sample t-test (Comparing Means)

- When to use: Numerical (continuous) outcome variable with two independent groups.
- Question answered: Is the average income different between men and women?
- Logic: The test calculates the difference in group means and adjusts for variability using the standard error, determining whether the observed gap is larger than what chance alone would produce.
- Formal name: Independent Two-Sample t-test
- Key assumption: The two groups are made up of different, unrelated individuals (independent samples).





Two-Sample z-test (Comparing Proportions)

- Used when the outcome is binary (yes/no) and we want to compare proportions between two independent groups.
- Tests whether the observed difference is larger than what chance alone would produce.

Key Details

- Two independent groups being compared
- Question: Is AI tool use different between adolescents and young adults?
- Example: Adolescents 45% vs. Young adults 50%
- Name: Two-proportion z-test
- Key idea: Is the difference larger than expected by chance?

One-Way ANOVA (Comparing Means Across Multiple Groups)

- Outcome variable is continuous (e.g., income, test score, hourly wage)
- Explanatory variable is categorical with three or more groups

Key Details

- Compares the variation between group means to the variation within groups
- Calculates an F-statistic:
  - Large F → group means are more different than expected by chance.
  - Small F → observed differences are likely due to random variation.
- A significant result indicates that at least one group mean differs from the others.
- ANOVA **does not identify which groups differ!**

Geographic Region	Mean wage
North	$15
South	$15
East	$18
West	$15

Chi-square Test

When to Use

- Use the chi-square test when both the outcome and explanatory variables are discrete.
- Data is organized in a contingency table, and the test evaluates whether distributions differ across groups.

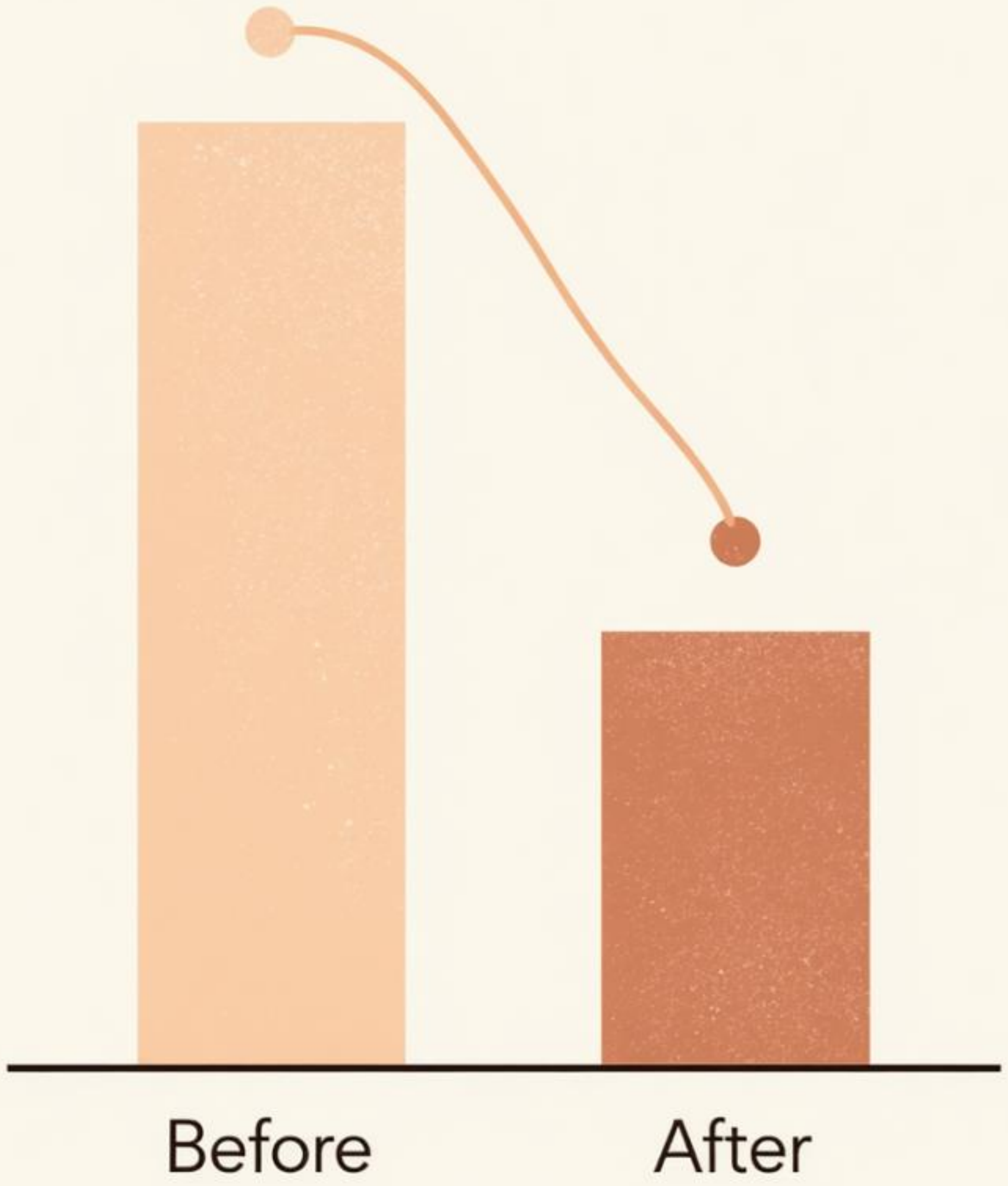
Key Concepts

- Both variables must be discrete (nominal or ordinal)
- Data arranged in a contingency table (rows × columns)
- Question: Is childcare provider type associated with race/ethnicity?
- Association exists if distributions differ meaningfully across groups. If the distributions are similar across all groups, the variables are likely not associated.

Race	Childcare center-based providers	Childcare home-based providers
White	70%	30%
Black	40%	60%
Asian	65%	35%

Dependent Samples (Paired Data)

- Compare the same individuals measured twice, or matched pairs (not two separate groups).
- Example: *test scores before vs. after tutoring; same child measured at two time points.*
- Method: compute individual change scores for each person, then analyze those changes.
- Why it works: each person serves as their own baseline, which isolates true change and removes between-person variability.



Key Concepts to Understand R Outputs

1. **Hypothesis:**
   - Null (H_0) assumes no difference exists.
   - Alternative (H_1) assumes a difference or relationship exists.

2. **P-value:**
   - Measures how consistent the sample results are with the null hypothesis.
   - Researchers often choose **5% cutoff** to accept/reject the null hypothesis ⟶ accepting about a 5% chance of incorrectly concluding there is a difference or relationship when none actually exists.
   - $p < 0.05$ → reject null (statistically significant).
   - $p \geq 0.05$ → no evidence to reject null.

Key Concepts to Understand R Outputs

3 **Confidence Interval (CI):**
Shows direction, evidence, size, and precision of the difference. If the difference is defined as
Difference = Group 1 Mean – Group 2 Mean, the confidence interval tells us:

Direction of the Difference

- Positive values → Group 1 has the higher average
- Negative values → Group 2 has the higher average
- Values near 0 → little or no difference between groups

Statistical Evidence

- Interval includes 0 → no clear evidence of a difference
- Interval excludes 0 → evidence of a difference between groups

Size of the Effect

- Larger values (farther from 0) → larger difference between groups
- Smaller values (closer to 0) → smaller difference between groups

Precision of the Estimate

- Narrow interval → more precise estimate
- Wide interval → less precise estimate

Statistical tests tell us if a difference is likely real, not how important or large that difference is.

How to Choose the Right Test

Question 1: What Is the Outcome?

- Numerical outcome → Use a t-test
- *Example: Comparing average income between men and women.*
- Categorical outcome → Use a z-test or chi-square test
- *Example: Comparing proportions or category distributions across groups.*

Question 2: How Are Groups Structured?

- Different people in each group → Use an independent samples test
- *Example: Men vs. women; treatment vs. control group.*
- Same people measured twice → Use a paired t-test
- *Example: Before vs. after an intervention; matched pairs.*

https://stats.oarc.ucla.edu/other/mult-pkg/whatstat/

Next Class

- Derived Variables
- Linear Regression
- Problem Set due Sunday at 11:59pm!

Questions?

CLASS 4: BIVARIATE DESCRIPTIVE STATISTICS AND BASIC STATISTICAL TESTS

Aida Pacheco-Applegate

Summer 2026

15-minute quiz +
5-minute break